LAIF

Life Advancement through Intelligent Forecasting

A Rwanda-led approach to earlier detection, connected care and safer medicine use. Anchored in national digital infrastructure, designed for clinical judgment, governed in Rwanda.

PROPOSAL FOR DISCUSSION WITH THE MINISTRY OF HEALTH, RWANDA

NATIONAL BLUEPRINT



4 provinces + City of Kigali · 30 districts

- **Predict**
- **Prevent**
- **Prosper**

STRATEGIC INTENT

Modular intelligence, anchored in Rwanda.

LAIF is an AI-driven, modular health-intelligence framework that helps people, clinicians and the wider health system act earlier — so a health signal turns into a completed step of care.

ALIGNMENT

+9.3%

GDP GROWTH

Targeted by Rwanda's NST2 — the framework LAIF is engineered to accelerate.

AUDIENCE 01

For people

Understand a recorded risk in clear language, know what to do next, and report what they take.

AUDIENCE 02

For health professionals

See authorized, relevant patient history and easily identify who needs review.

AUDIENCE 03

For the health system

Measure whether screening leads to referral, care and continued follow-up.

CONTEXT · WHY THIS MATTERS TO RWANDA

A strong foundation. The next chance is connected prevention.

Rwanda has invested in screening, identity and a national health intelligence layer. LAIF closes the loop between them.

01

STRATEGIC ALIGNMENT

Health Sector Strategic Plan V prioritizes coverage, quality, workforce, research and digitalisation through 2028/29.

02

NATIONAL INTELLIGENCE

The National Health Intelligence Center already brings health evidence into one national layer. LAIF links action to it.

03

NCD BASELINE

The 2024–25 NCD report already measures blood pressure, blood sugar and BMI. LAIF tracks what happens next.

04

LAIF CONTRIBUTION

Help track what happens after screening: review, referral, care and continued follow-up — then measure it.

> **LAIF is designed with Rwanda, governed in Rwanda, scaled with Rwanda.**

THE PROBLEM WE WANT TO SOLVE

Three critical frictions in current care.

“

A health risk may be identified without a complete path to action.

The next three slides describe the everyday frictions that keep good intentions from becoming completed care.

01

Fragmented data silos

Clinical records, diagnostic labs, pharmacy logs, and community wellness indicators sit disjointed. Continuity of care is interrupted, and patterns invisible in one file remain invisible everywhere.

02

The identification paradox

Many legacy systems lack reliable identity mapping, so individuals re-register at each facility. Demographic tracing becomes guesswork and consent becomes harder to enforce.

03

Reactive public health models

Policy is locked into backward-looking reports. By the time a district sees an outbreak curve, the moment for prevention has already passed. Action comes after the fact, not before it.

Response lag

WHAT LAIF IS

One platform. Three connected experiences.

A single intelligence layer opens a different window for each audience — and the same health signal flows through all three.

CITIZEN VIEW

Understand the next step.

Screenings in clear language, agreed reminders, and simple reporting of medicines, allergies or symptoms.

APPLICATION

Predictive Health Wallet

Mobile dashboard linked to e-ID: trajectory forecasts, nudges, daily risk indicators.

HEALTH PROFESSIONAL VIEW

Prioritize, then close the loop.

Authorized history, clinically-reviewed prompts, visible referrals — dismiss or correct with a recorded reason.

APPLICATION

Clinical Forecast Portal

District EHR workspace: chronic risk, diabetic markers and medication conflicts early.

MINISTRY & PROGRAMMES

Measure what's working.

Follow-up completion, district differences, care gaps, safety issues and worker burden — aligned with NHIC reporting.

APPLICATION

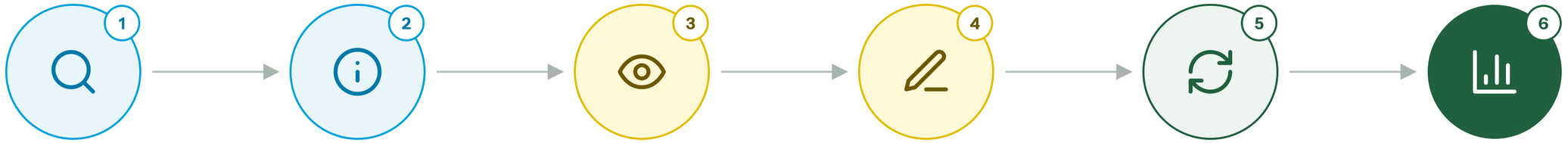
National Policy Sandbox

MoH population dashboard: regional heatmaps to direct medicines, clinics and resources.

THE CARE JOURNEY

From a health signal to a completed follow-up.

Six repeatable steps — each closing today, ready to begin tomorrow.



Identify

A screening or relevant health observation is recorded once, in the agreed place.

Understand

The person sees the result and the appropriate next step, in language they understand.

Review

An authorized professional reviews relevant information and the prompt behind it.

Act

Advice, testing, treatment or referral is recorded — or dismissed with a reason.

Follow up

The care team — and the system — check whether the next step actually occurred.

Learn

Approved aggregate measures show where the pathway works — and where it needs improvement.

Measures improve the next cycle

JOURNEY 01 · CITIZEN

A citizen’s prevention journey.

Understand risks. Take action. Follow your progress.

Steps 1–3 **Understand**

Open a secure health profile, add screening results and history, and see clearly what needs attention.

Steps 4–5 **Act**

Agree a prevention plan with a health professional, then follow agreed daily reminders.

Steps 6–7 **Follow up**

Repeat checks are compared with earlier records; progress and the next review stay visible.

Illustrative and anonymised — no real patient is referenced.



JOURNEY 02 · HOSPITAL & CLINICIAN

A connected hospital visit.

Proposed workflow — subject to Rwanda's approvals and clinical governance.

Steps 1–3 — **Identify**

The patient arrives, presents a National ID, and an approved biometric check confirms identity.

Steps 4–5 — **Care**

The physician reviews authorized history, then examines and treats — the clinician decides.

Steps 6–7 — **Record**

X-rays, scans and reports are uploaded; the prescription and visit are saved to the patient record.

National ID links an authorized record; clinical data remains protected and every access is logged.



JOURNEY 03 · NATIONAL HEALTH LEADERSHIP

From protected data to measurable action.

Ministry and programme leads see aggregate, privacy-protected evidence — and act on it.

Steps 1–2

See

Role-based secure sign-in opens the national picture, with trends down to district level.

Steps 3–4

Decide

Detect district gaps — such as missed hypertension follow-up — then validate data quality and context.

Steps 5–7

Act & measure

Plan the intervention, mobilise community health workers, and measure referral completion and follow-up.

Dashboard views are illustrative mock-ups; figures shown are not Ministry data.



MODULAR STACK INTEGRATION

A six-layer architecture, ordered from foundation to citizen.

Level 5	**Experience layers**	Citizen mobile + hospital portals
Level 4	**AI intelligence core**	Predictive indices 3–12 months ahead
Level 3	**Canonical FHIR model**	Standardised demographic data
Level 2	**Identity mapping (NIDA e-ID)**	API validation handshake
Level 1	**Interoperability bridge**	Real-time adapters & EHR ingest
Level 0	**Sovereign infrastructure host**	Regional cloud & local residency

ARCHITECTURAL ALIGNMENT

Every layer maps naturally to Rwandan digital infrastructure.

Legacy systems do not need to be rebuilt. LAIF acts as an ingestion translation bridge, keeping primary data inside national boundaries while exposing hooked-in intelligence upward.

DATA PROTECTION HANDSHAKE

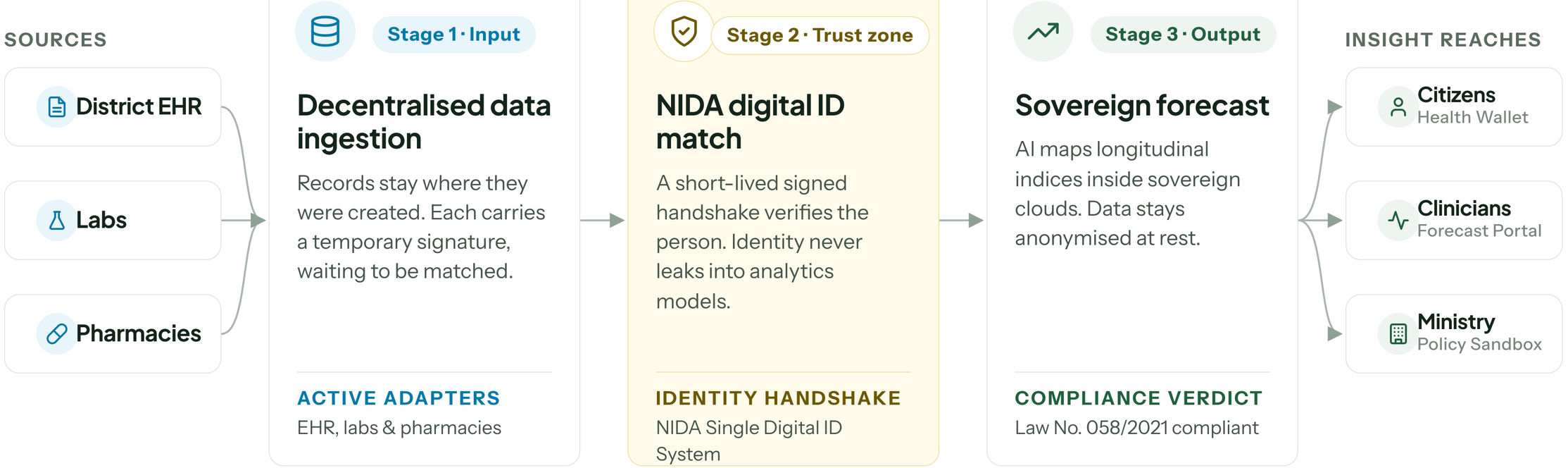
NIDA API queries use ephemeral tokenisation endpoints — records match, personally identifiable information never caches on analytics models.

DATA FLOW & PRIVACY ARCHITECTURE

An end-to-end pipeline, anchored on NIDA trust.

Three stages. Records stay close to the citizen. Insight reaches the people who need it.

SOVEREIGN BOUNDARY · RECORDS, IDENTITY AND FORECASTS STAY IN RWANDA



Strict compliance. Sovereign trust by design.

NATIONAL TRUST PILLAR

Rwanda Law No. 058/2021 alignment.

48h

STRICT BREACH DISCLOSURE

Controllers notify NCSA within 48 hours; processors notify the controller within 48 hours. A written report follows within 72 hours.

- **Stored in Rwanda by default.** Storage abroad needs an NCSA registration certificate (Art. 50); LAIF commits to in-country hosting.
- **Data protection officer.** Required for corporate bodies and large-scale or sensitive processing (Art. 40); logs record access, alteration and combination.
- **Decentralised handshakes.** Health files link to NIDA via temporary, signed tokens.

GLOBAL HARMONIZATION

Converging with global standards.



LAIF upholds data portability, the right to erasure, full transparency controls and end-to-end encryption at rest and in transit. Bar lengths are illustrative — actual controls are maintained by NCSA.

Clinical trust comes before automated prediction.

A PRINCIPLE BEFORE A FEATURE

A clinician must always be able to challenge an output.

05 No AI independently diagnoses, prescribes, or denies care.



01 Transparent rules first **02** Suitability, validation, oversight **03** Audited for false alarms & missed cases **04** Always open to challenge **05** Humans decide

PILOT USE CASE · 01

Make screening useful by connecting it to care.

NCD follow-up — Rwanda's 2024–25 NCD report already measures the basics. LAIF tracks what happens after the cuff is removed.

SCOPE OF THE PILOT

- Adults screened for elevated blood pressure or blood sugar through participating services
- Record or receive screening outcomes and apply Ministry-approved clinical review criteria
- Assign follow-up actions; track assessment and referral; send agreed reminders
- Measure time from screening to review and the percentage of completed follow-ups

PILOT OBJECTIVE

Test whether connected prevention helps — without imposing an unmanageable health-worker burden.

CONCRETE PATHWAY

From a cuff to a closed review.

A short, governed care pathway. Ministry-approved rules. Each step recorded, each dismissal justified, each reminder explained.

WHAT THE PILOT DELIBERATELY EXCLUDES

- No clinical autonomy transfers.
- No automated prescribing.
- No AI-only diagnosis or denial of care.
- No data stored outside Rwanda.

PILOT USE CASE · 02

Safer medicine use, with the clinician in charge.

Two careful domains — antibiotics, and pain medicines. Patient reports and provider records meet on one timeline.

DOMAIN A

Why antibiotic use matters.

Antibiotics work against bacterial infections — not against feeling unwell. The bacteria, not the patient, become resistant.

- Inappropriate use fuels antimicrobial resistance — a higher dose is not the answer
- LAIF supports patient education, medicine histories and clinical review
- Aligned with Rwanda's AMR National Action Plan 2025–2029

DOMAIN B

Painkillers need a different approach.

Risk depends on the molecule, the dose, the duration and the person. Treating every medicine as the same is a mistake.

- Patients can report non-prescription medicines they take at home
- Clinicians see recorded medicines and relevant conditions on one page
- Repeated or persistent pain triggers a professional review, never more pills

LAIF never prescribes, never raises doses, and never instructs a patient to stop a medicine. The clinician decides; LAIF shows the context.

A personal wellness score, with four ways to use it.

REAL-TIME PERSONAL INDEX

LAIF Wellness Score

A privacy-preserving, cryptographically signed index scoring lifestyle habits, metabolic metrics and chronic risks, securely synced to the citizen's e-ID.



Fully compliant with Law No. 058/2021

INSURANCE · FINANCIAL

Insurance & underwriting

Underwriters can offer discounted premiums to citizens showing sustained healthy-scoring trends.

MUTUELLE · PUBLIC HEALTH

Mutuelle optimisation

Resources reach regions with low geographic indices, dynamically targeting preventive campaigns.

MICROFINANCE · CREDIT

Microfinance multipliers

Continuity-of-wellness records provide supplementary reliability for unbanked citizens seeking micro-credit.

ECOSYSTEM · COMMERCE

Ecosystem rewards

Local pharmacies, gyms and shops offer loyalty credits tied to continuous healthy-score increments.

PILOT EVALUATION FRAMEWORK

Six ways we will know whether LAIF helped.

Baselines, targets and methods are agreed before launch. We do not claim reduced resistance without evidence.

KPI 01 · PATHWAY COMPLETION

Follow-up completed

Did more people receive the assessment they were flagged for, end to end?

KPI 02 · PATHWAY TEMPO

Time to clinical review

Median days from screening event to professional review and referral completion.

KPI 03 · HISTORY QUALITY

Medicine-history completeness

Did clinicians see a more useful account of patient-reported and recorded medicines?

KPI 04 · WORKER EXPERIENCE

Health-worker workload

Minutes per patient-day indexed against baseline. Burden must not rise to maintain benefit.

KPI 05 · CITIZEN CLARITY

Patient understanding

Did the citizen understand their screening, the next step, and how their data is being used?

KPI 06 · SAFETY & FAIRNESS

Reliability of prompts & ID

False alarms, missed cases, identity mismatches — all reviewed on a public dashboard.

COMPARATIVE CONTEXT

Rwanda is unusually well-positioned to lead this work.

Few African contexts combine national digital ID, a community-health backbone and an enforced data-protection regime. LAIF turns those conditions into working prevention.

DIGITAL IDENTITY

NIDA digital ID, launched at Expo 2025 in Gikondo.

A national identity layer most peer systems cannot yet count as settled.

COMMUNITY BACKBONE

A national community-health-worker network.

Coherence at village level that few peers match at the same depth.

MUTUELLE DE SANTÉ

Near-universal community health insurance.

A system that can route prevention spend at district granularity.

LEGAL & INTELLIGENCE LAYER

NCSA + NST2 + National Health Intelligence Center.

A regulator that exists, a strategy that names prevention, a national centre ready to receive.

WHAT THIS MEANS FOR LAIF

The integration problem is already solved. The clinical problem is now the priority.

Where peers often spend their first three years wiring identity, payment rails and regulators, Rwanda begins the predictive clinical work from a higher floor.

A NOTE ON HONESTY

Where this deck references peers it does so directionally. National capability rankings vary by methodology and update. Rwanda's standing is stated factually, not as competitive boast.

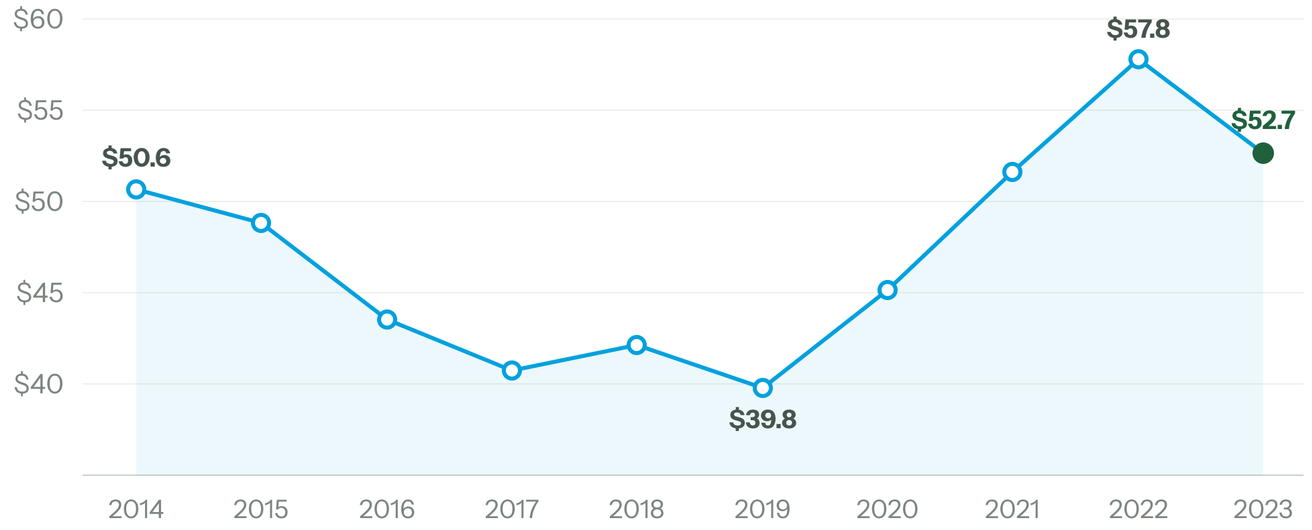
HEALTHCARE EXPENDITURE

Tight budgets, rising NCDs: prevention pays.

Rwanda spends about US$53 per person on health, 37% of it externally financed — while NCDs now cause nearly half of all deaths.

CURRENT HEALTH EXPENDITURE PER CAPITA · 2014 → 2023

US$ per person, current prices



Source: WHO Global Health Expenditure Database via World Bank (SH.XPD.CHEX.PC.CD). 2023 = 5.1% of GDP.

HEALTH SHARE OF NATIONAL BUDGET

7.8% FY2022/23, against the **15%** Abuja target

NCD SHARE OF DEATHS

47.7% of all deaths in 2024, up from 44% (WHO, 2016)

RETURN ON PREVENTION · WHO BENCHMARK

7× return from NCD “Best Buys” at about US$3 per person a year (LMICs)

STRATEGIC ROLLOUT SCHEDULE

Implementation roadmap & milestones.

Twelve months, three phases. Each phase has a clear outcome the Ministry can review before the next begins.

M1 M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 M12

PHASE 01 · MONTHS 1–3
Sovereign sandbox

Technical due-diligence, joint governance drafts and sanitised testing with hospital dummy cohorts.

- API mapping with NIDA's digital ID system
- MoH + NCSA governance guidelines
- Sanitised testing on dummy cohorts

PHASE 02 · MONTHS 4–8
Clinical pilot

Deploying in-country gateways inside regional cloud nodes. Diagnostic data matched for initial district hospital groups.

- Regional gateway deployment complete
- District hospital clinical data paths
- Clinician UI feedback & evaluations

PHASE 03 · MONTHS 9–12
Civic scaling

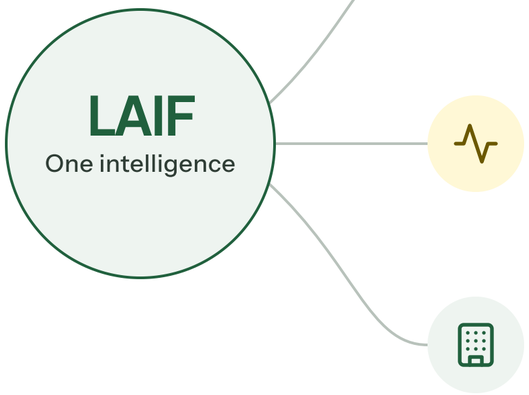
Personalised prediction rolled across Rwanda's national ecosystem — extending into King Faisal Hospital and rural community networks.

- Mobile health wallet in national ecosystem
- Network extended to rural health posts
- First national predictive heatmap

◇ **Ministry review gate — each phase begins only after the previous one is reviewed and explicitly authorised.**

NATIONAL VALUE & PARTNERSHIP

Three sectors share the value. Designed in Rwanda. Governed in Rwanda.



CITIZENS & COMMUNITIES

Personal, decentralised custody of one's clinical timeline.

Daily nudges & tailored warnings · Visibility on access to personal data

CLINICIANS & HEALTH SYSTEMS

Predictive, real-time longitudinal history.

Triage lists ranked by reason · Verified drug-conflict histories

STATE & MINISTRY

Acceleration under NST2.

Live regional heatmaps · Direct vehicles for NST2 targets

PROPOSED PARTNERSHIP MODEL

Designed with Rwanda, governed in Rwanda.

MINISTRY LEADS

- Priorities
- Clinical protocols
- Locations
- Approvals
- Evaluation

OUR TEAM CONTRIBUTES

- Product design
- Prototype
- Implementation support
- Training
- Reporting tools

Long-term operation remains Rwandan.

WHAT WE ASK THE MINISTRY TO ANCHOR

1. Counterparts in clinical, public health, digital health & data governance
2. Agreement on a formal LAIF pilot protocol
3. Joint scoping of safeguards, budget, timeline, success criteria
4. A first joint working session — **the immediate next step**

PREDICT · PREVENT · PROSPER

Murakoze Cyane.

Shaping a healthier, more sovereign national future — one health signal, one completed step of care at a time.

When a risk is detected, a person should understand the next step.

When a professional provides care, relevant information should be available safely.

When a programme invests in prevention, it should measure what happens next.

THE NEXT STEP

A joint working session, soon.

One pilot, agreed protocol, real measures.